FILTERED LOCALIZATION SYSTEMS OF DISTINGUISHED TRIANGLES AND COFINALITY OF COMPONENT FUNCTORS

THE STACKS PROJECT AUTHORS

Editorial note. References to standard Categories inputs display their stable Stacks tags. The corresponding permitted inputs are explicitly stated in task metadata.

Editorial note. Triangle compatibility means MS5 (shift stability of S) and MS6 (completion of two S-maps to a third S-map in a morphism of distinguished triangles). These are hypotheses, not unproved solution lemmas.

Editorial scope. One original lemma. Standard calculus of fractions is explicitly allowed; all source-local triangle coherence remains in original proof.

Allowed standard prerequisites (editorial).

- Calculus of fractions for a saturated multiplicative system. Ore completion, common-denominator equalization, filtered categories X/S, and equality in localization after postcomposition by an S-map. Saturation says any map inverted by localization lies in S.. Categories tags05Q0 and05Q9; these are standard calculus-of-fractions inputs, not triangle coherence.
- Pre-triangulated rotation and compatibility MS5/MS6. S is stable under shift; every commutative square between first two entries of distinguished triangles with both maps in S admits a third S-map completing the triangle morphism.. derived.tex1156-1181
- Filtered categories and cofinal functors. Filtered means nonempty, common upper targets and equalization of parallel arrows; cofinal means each target receives an arrow and the corresponding comma categories are connected.. Categories tags002V and04E6

Source reference. Original lemma 05Q9: focused-seeds/ACL/sources/stacks-snapshot-20261011/categories.tex, source lines 4795–4811. Explicit permitted standard input; the target triangle-coherence construction is not supplied by this reference. Display number is stable Stacks tag, not chapter numbering.

Source reference. Original definition 002V: focused-seeds/ACL/sources/stacks-snapshot-20261011/categories.tex, source lines 2130–2147. Explicit permitted standard input; the target triangle-coherence construction is not supplied by this reference. Display number is stable Stacks tag, not chapter numbering.

Source reference. Original definition 04E6: focused-seeds/ACL/sources/stacks-snapshot-20261011/categories.tex, source lines 1725–1751. Explicit permitted standard input; the target triangle-coherence construction is not supplied by this reference. Display number is stable Stacks tag, not chapter numbering.

Source reference. Original section 05Q0: focused-seeds/ACL/sources/stacks-snapshot-20261011/categories.tex, source lines 4362–4399. Explicit permitted standard input; the target triangle-coherence construction is not supplied by this reference. Display number is stable Stacks tag, not chapter numbering.

Date: stacks-project:derived.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\mathcal{D}$ be a pre-triangulated category. Let S be a saturated multiplicative system in $\mathcal{D}$ that is compatible with the triangulated structure. Let (X, Y, Z, f, g, h) be a distinguished triangle in $\mathcal{D}$. Consider the category of morphisms of triangles*

$$\mathcal{I} = \{(s, s', s'') : (X, Y, Z, f, g, h) \to (X', Y', Z', f', g', h') \mid s, s', s'' \in S\}$$

Then $\mathcal{I}$ is a filtered category and the functors $\mathcal{I} \to X/S$, $\mathcal{I} \to Y/S$, and $\mathcal{I} \to Z/S$ are cofinal.

Proof. We strongly suggest the reader skip the proof of this lemma and instead work it out on a napkin.

The first remark is that using rotation of distinguished triangles (TR2) gives an equivalence of categories between $\mathcal{I}$ and the corresponding category for the distinguished triangle $(Y, Z, X[1], g, h, -f[1])$. Using this we see for example that if we prove the functor $\mathcal{I} \to X/S$ is cofinal, then the same thing is true for the functors $\mathcal{I} \to Y/S$ and $\mathcal{I} \to Z/S$.

Note that if $s : X \to X'$ is a morphism of S, then using MS2 we can find $s' : Y \to Y'$ and $f' : X' \to Y'$ such that $f' \circ s = s' \circ f$, whereupon we can use MS6 to complete this into an object of $\mathcal{I}$. Hence the functor $\mathcal{I} \to X/S$ is surjective on objects. Using rotation as above this implies the same thing is true for the functors $\mathcal{I} \to Y/S$ and $\mathcal{I} \to Z/S$.

Suppose given objects $s_1 : X \to X_1$ and $s_2 : X \to X_2$ in X/S and a morphism $a : X_1 \to X_2$ in X/S. Since S is saturated, we see that $a \in S$, see Categories, Lemma 05Q9. By the argument of the previous paragraph we can complete $s_1 : X \to X_1$ to an object $(s_1, s_1', s_1'') : (X, Y, Z, f, g, h) \to (X_1, Y_1, Z_1, f_1, g_1, h_1)$ in $\mathcal{I}$. Then we can repeat and find $(a, b, c) : (X_1, Y_1, Z_1, f_1, g_1, h_1) \to (X_2, Y_2, Z_2, f_2, g_2, h_2)$ with $a, b, c \in S$ completing the given $a : X_1 \to X_2$. But then (a, b, c) is a morphism in $\mathcal{I}$. In this way we conclude that the functor $\mathcal{I} \to X/S$ is also surjective on arrows. Using rotation as above, this implies the same thing is true for the functors $\mathcal{I} \to Y/S$ and $\mathcal{I} \to Z/S$.

The category $\mathcal{I}$ is nonempty as the identity provides an object. This proves the condition (1) of the definition of a filtered category, see Categories, Definition 002V.

We check condition (2) of Categories, Definition 002V for the category $\mathcal{I}$. Suppose given objects $(s_1, s_1', s_1'') : (X, Y, Z, f, g, h) \to (X_1, Y_1, Z_1, f_1, g_1, h_1)$ and $(s_2, s_2', s_2'') : (X, Y, Z, f, g, h) \to (X_2, Y_2, Z_2, f_2, g_2, h_2)$ in $\mathcal{I}$. We want to find an object of $\mathcal{I}$ which is the target of an arrow from both $(X_1, Y_1, Z_1, f_1, g_1, h_1)$ and $(X_2, Y_2, Z_2, f_2, g_2, h_2)$. By Categories, Remark 05Q0 the categories X/S, Y/S, Z/S are filtered. Thus we can find $X \to X_3$ in X/S and morphisms $s : X_2 \to X_3$ and $a : X_1 \to X_3$. By the above we can find a morphism $(s, s', s'') : (X_2, Y_2, Z_2, f_2, g_2, h_2) \to (X_3, Y_3, Z_3, f_3, g_3, h_3)$ with $s', s'' \in S$. After replacing (X_2, Y_2, Z_2) by (X_3, Y_3, Z_3) we may assume that there exists a morphism $a : X_1 \to X_2$ in X/S. Repeating the argument for Y and Z (by rotating as above) we may assume there is a morphism $a : X_1 \to X_2$ in X/S, $b : Y_1 \to Y_2$ in Y/S, and $c : Z_1 \to Z_2$ in Z/S. However, these morphisms do not necessarily give rise to a morphism of distinguished triangles. On the other hand, the necessary diagrams do commute in $S^{-1}\mathcal{D}$. Hence we see (for example) that there exists a morphism $s_2' : Y_2 \to Y_3$ in S such that $s_2' \circ f_2 \circ a = s_2' \circ b \circ f_1$. Another replacement of (X_2, Y_2, Z_2) as above then gets us to the situation where $f_2 \circ a = b \circ f_1$. Rotating and applying the same argument two more times we see that we may assume (a, b, c) is a morphism of triangles. This proves condition (2).

Next we check condition (3) of Categories, Definition 002V. Suppose $(s_1, s_1', s_1'') : (X, Y, Z) \to (X_1, Y_1, Z_1)$ and $(s_2, s_2', s_2'') : (X, Y, Z) \to (X_2, Y_2, Z_2)$ are objects of $\mathcal{I}$, and suppose $(a, b, c), (a', b', c')$ are two morphisms between them. Since $a \circ s_1 = a' \circ s_1$ there exists a morphism $s_3 : X_2 \to X_3$ such that $s_3 \circ a = s_3 \circ a'$. Using the surjectivity statement we can complete this to a morphism of triangles $(s_3, s_3', s_3'') : (X_2, Y_2, Z_2) \to (X_3, Y_3, Z_3)$ with $s_3, s_3', s_3'' \in S$. Thus $(s_3 \circ s_2, s_3' \circ s_2', s_3'' \circ s_2'') : (X, Y, Z) \to (X_3, Y_3, Z_3)$ is also an object of $\mathcal{I}$ and after composing the maps $(a, b, c), (a', b', c')$ with (s_3, s_3', s_3'') we obtain $a = a'$. By rotating we may do the same to get $b = b'$ and $c = c'$.

Finally, we check that $\mathcal{I} \to X/S$ is cofinal, see Categories, Definition 04E6. The first condition is true as the functor is surjective. Suppose that we have an object $s : X \to X'$ in X/S and two objects $(s_1, s_1', s_1'') : (X, Y, Z, f, g, h) \to (X_1, Y_1, Z_1, f_1, g_1, h_1)$ and $(s_2, s_2', s_2'') : (X, Y, Z, f, g, h) \to (X_2, Y_2, Z_2, f_2, g_2, h_2)$ in $\mathcal{I}$ as well as morphisms $t_1 : X' \to X_1$ and $t_2 : X' \to X_2$ in X/S. By property (2) of $\mathcal{I}$ proved above we can find morphisms $(s_3, s_3', s_3'') : (X_1, Y_1, Z_1, f_1, g_1, h_1) \to (X_3, Y_3, Z_3, f_3, g_3, h_3)$ and $(s_4, s_4', s_4'') : (X_2, Y_2, Z_2, f_2, g_2, h_2) \to (X_3, Y_3, Z_3, f_3, g_3, h_3)$ in $\mathcal{I}$. We would be done if the compositions $X' \to X_1 \to X_3$ and $X' \to X_2 \to X_3$ were equal (see displayed equation in Categories, Definition 04E6). If not, then, because X/S is filtered, we can choose a morphism $X_3 \to X_4$ in X/S such that the compositions $X' \to X_1 \to X_3 \to X_4$ and $X' \to X_2 \to X_3 \to X_4$ are equal. Then we finally complete $X_3 \to X_4$ to a morphism $(X_3, Y_3, Z_3) \to (X_4, Y_4, Z_4)$ in $\mathcal{I}$ and compose with that morphism to see that the result is true. $\square$